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This CV is presented for the IEEE Transactions on Information, Forensics, and Security journal’s associate editor nomination.

Working Experience

Researcher

Department of Computer Science and Technology, Nanjing University, China 2019.01-2023.06

Associate Professor

Software Institute, Nanjing University, China 2023.07-now

Education Experience

Ph.D. of Computer Science

Computer Science and Technology, Nanjing University, China 2013.09-2018.12

Visiting Scholar

School of Computer Science, William & Mary, USA 2016.10-2018.02

Bachelor of Computer Science

Computer Science and Technology, Nanjing University, China 2009.09-2013.06

Recent Research Work

Symbols # and * indicate student authors and faculty authors, respectively.

Auditing Data Provenance in LLM Fine-tuning via Intrinsic Distributional Fingerprints

Zirui Huang#, Yunlong Mao*, Wei Tong*, Tingting Wu, Xin Ge, and Sheng Zhong* CCS 2026

- A new dataset fingerprint technique for the provenance of LLM fine-tuning data.

Fend for Yourself! Backdoor Purification in Federated Graph Learning with an Evolving Knowledge Anchor

Chengcheng Zhu#, Yunlong Mao*, Jiale Zhang, Bosen Rao, and Sheng Zhong* Security 2026

- Mitigating backdoor attacks on federated learning in a distributed and independent manner.

ObfusLM: Privacy-preserving Language Model Service against Embedding Inversion Attacks

Yu Lin, Ruining Yang, Yunlong Mao*, Qizhi Zhang, Jue Hong, Quanwei Cai, Ye Wu, Huiqi Liu, Zhiyu Chen, Bing Duan, and Sheng Zhong* ACL 2026

- A defense solution for embedding inversion attacks with lightweight encryption.

FL-MPS: Secure Federated LLM Fine-Tuning via Adaptive Parameter Split and Efficient Negotiation

Yuejuan Zhu#, Yunlong Mao*, Guanzhi Qiu#, Shijie Ding#, Wenjun Qian, Sheng Zhong* IEEE TDSC 2026

- Fast and secure aggregation solution for privacy-preserving federated learning.

SPA: Stealthy and Persistent Backdoor Attacks in Federated Learning via Feature-Space Alignment

Chengcheng Zhu#, Ye Li, Bosen Rao, Yunlong Mao*, Jiale Zhang, Tingting Wu, Xin Ge, and Sheng Zhong* IEEE TIFS 2026

- Feature alignment-based backdoor attack in federated learning.

Designing Wallet-Based User Intervention for Approval Phishing Mitigation

Maggie Yongqi Guan, Yuqi Xu#, Yunlong Mao*, Wei Tong*, Xiaobo Zhou and Kanye Ye Wang Security 2026

- Improvement of web3 wallet design to mitigate approval phishing attacks.

Liquidity Mining as an Attack Surface: Incentive-Induced Liquidity Attacks in Concentrated Liquidity Market Makers

Sinong Lu, Chon Kit Lao, Yunlong Mao*, Xiaobo Zhou, Ruizhe Jia, and Kanye Ye Wang Security 2026

- A new liquidity mining attack study of market makers.

Designing Interface Interventions Against Address Poisoning Attacks in Web3 Wallets Yuqi Xu#, Maggie Yongqi Guan, Yunlong Mao* , Xin Yi, Wei Tong*, Sheng Zhong*, Xiaobo Zhou, and Kanye Ye Wang ○ Improvement of web3 wallet design to mitigate address phishing attacks.	UIST 2026
Counselors' Experiences of Privacy and Care in Anonymous Gambling Helplines Molly Zhuangtong Huang, Jingjie Li, Sihong Teng, Yunlong Mao* , Sheng Zhong*, Xiaobo Zhou and Kanye Ye Wang ○ An experience study of user privacy and healthcare in anonymous gambling.	UbiComp 2026
COVERT: Privacy-Preserving Covariant Obfuscation for VLMaaS via Exact Reparameterization and Tailored Tuning Wenqiang Ruan, Zirui Huang#, Yu Lin, Qizhi Zhang, Yunlong Mao* , Quanwei Cai, Jue Hong, Sheng Zhong*, and Ye Wu ○ Secure multimodal LLM inference via model weight obfuscation.	ECCV 2026
Preventing Dataset Misuse in Stable Diffusion for Multi-User Sharing Yubo Zhu#, Xinyue Zhao#, Zijian Wang#, Wei Tong*, Yunlong Mao* , and Sheng Zhong* ○ Ensuring data security in stable diffusion models with user-sharing interactions.	ICME 2026
SAP: Privacy-Preserving Fine-Tuning on Language Models with Split-and-Privatize Framework Xicong Shen, Yang Liu, Yi Liu, Peiran Wang, Huiqi Liu, Jue Hong, Bing Duan, Zirui Huang#, Yunlong Mao* , Ye Wu, and Sheng Zhong* ○ Preserving data privacy for LLM fine-tuning in a split learning framework.	IJCAI 2025
LT-OAQ: Learnable Threshold Based Outlier-aware Quantization And Its Energy Efficient Accelerator For Low-precision On-chip Training Qinkai Xu#, Yijin Liu, Yuan Meng, Yang Chen, Yunlong Mao* , Li Li*, and Yuxiang Fu* ○ A robust and efficient model weight quantization method for on-chip model training.	DATE 2025
CrossNet: A Low-Latency MLaaS Framework for Privacy-Preserving Neural Network Inference on Resource-Limited Devices Yu Lin#, Tianling Zhang#, Yunlong Mao* , Sheng Zhong* ○ A secure and efficient DNN inference solution for lightweight devices.	TDSC 2024
Secure Model Aggregation against Poisoning Attacks for Cross-silo Federated Learning with Robustness and Fairness Yunlong Mao* , Zhujiang Ye#, Xinyu Yuan#, Sheng Zhong* ○ A secure and fair model aggregation solution for heterogeneous federated learning.	TIFS 2024
Solution Probing Attack Against Coin Mixing Based Privacy-Preserving Crowdsourcing Platforms Yunlong Mao* , Ziqin Dang#, Heng Wang#, Yuan Zhang*, Sheng Zhong* ○ A malicious user attack on digital coin mixing in anonymous crowdsourcing tasks.	TDSC 2024
UBA-Inf: Unlearning Activated Backdoor Attack with Influence-Driven Camouflage Zirui Huang#, Yunlong Mao* , Sheng Zhong* ○ Implant and activate backdoor using machine unlearning algorithms.	Security 2024

Recent Academic Service

Technical Program Committee 2027

- USENIX Security
- SaTML

Technical Program Committee 2026

- ACM CCS
- AAAI
- IEEE Globecom

Technical Program Committee 2025

- NeurIPS
- AAAI
- IEEE Globecom
- IEEE HPCC

Journal Reviewer 2025-2027, finished more than 50 reviewer jobs

- IEEE Transactions on Information Forensics and Security
- IEEE Transactions on Dependable and Secure Computing
- IEEE Transactions on Mobile Computing
- IEEE Transactions on Vehicular Technology
- IEEE/ACM Transactions on Networking, Computers & Security
- IEEE Internet of Things Journal
- ACM Transactions on Privacy and Security
- ...